

Setting Up pass: a Unix Password Manager

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Figure 1: A small wooden box with a brass clasp, lid slightly ajar to reveal index tabs within, a brass key resting beside it on pale stone.

A private, organised credential store: accessible only by key, each entry discrete and retrievable.

1 Introduction

I did not really appreciate how little control I had over my passwords until I tried to export them. I had accumulated over 600 credentials across a combination of 1Password, iCloud Keychain, and Chrome’s built-in password manager: three separate systems with no coherent policy, no consistent backup strategy, and no way to inspect the storage format without paying for a subscription.

The turning point came when I discovered **pass**, the standard Unix password manager. It stores every password as a GPG-encrypted file in a plain directory tree. The store lives in `~/.password-store/` (a non-cloud-mounted path, outside any Dropbox or iCloud hierarchy), can be tracked in a private git repository, and requires no proprietary client to read: any machine with GPG installed can decrypt an entry. The architecture is almost comically simple, which is precisely what makes it trustworthy.

This post documents the setup procedure: generating a GPG key, installing `pass`, initialising the store, and migrating a 1Password vault. It also records the incident that followed the migration: accidentally committing `.password-store` to a public dotfiles repository. That incident was a symptom of a prior architectural error: the store had been placed under a Dropbox-synced path, which is wrong on two grounds. First, a git repository in a cloud-mounted directory is subject to index corruption from sync races (a point developed in [post 64](#)). Second, an encrypted credential store belongs to no one's sync pipeline but your own. This post corrects both errors.

1.1 Motivations

- Browser-based password managers (Chrome, iCloud Keychain) are opaque: the storage format is proprietary, the export options are limited, and the data is commingled with browser state.
- Storing credentials in a paid subscription service creates a single point of failure: if the service changes its pricing or closes, migrating 600 entries becomes an urgent, unplanned task.
- A plain-file store tracked with git provides a full audit trail: every addition, deletion, and edit is a commit with a timestamp.
- GPG encryption is the same mechanism that protects software release signatures; using it for daily credential storage is a concrete way to build familiarity with the toolchain.
- A terminal-native workflow integrates naturally with the rest of a command-line research environment without requiring a browser extension or GUI application.

1.2 Objectives

1. Generate a GPG key and confirm it is stored in the system keyring.
2. Install `pass` and initialise a new password store keyed to that GPG identity.
3. Insert at least one credential by hand and retrieve it via clipboard copy.
4. Migrate an existing 1Password CSV export into the store using the `1password2pass.rb` conversion script.

This post documents the author's own learning process. Errors spotted or better approaches are welcome in the comment thread below.



Figure 2: Workspace ambiance: a desk with a terminal window open, a small keyring visible to the side.

2 What is pass?

`pass` is a shell script that wraps GPG file encryption and presents a password store as a directory tree of `.gpg` files. Think of it as a filing cabinet where every drawer is sealed with your GPG key: only someone who holds the corresponding private key can open a drawer, and the cabinet itself is just a folder on your hard drive. A concrete example: `pass insert email/gmail` encrypts whatever you type at the prompt and writes the ciphertext to `~/.password-store/email/gmail.gpg`; `pass -c email/gmail` decrypts that file and copies the first line (the password) to the clipboard for 45 seconds before clearing it.

3 Prerequisites

This post assumes:

- **Operating system:** macOS 13+ (Apple Silicon or x86_64) or a Linux distribution with bash 5+
- **Already installed:** Homebrew (macOS); GPG is installed as a dependency automatically
- **Background knowledge:** comfort running shell commands and editing dotfiles; basic understanding of public-key cryptography is helpful but not required
- **1Password export (optional):** a CSV export of an existing vault, needed only for the migration section

- **Time required:** approximately 20 minutes for install, GPG key generation, and a first smoke test; the migration step depends on vault size

On Linux, substitute `apt install pass gnupg` for the Homebrew commands below.

4 Installation

On macOS, Homebrew installs `pass` and its GPG dependencies in a single command. The `pinentry-mac` package provides the graphical passphrase dialog that macOS requires for GPG operations outside a terminal session.

```
brew install pass gnupg pinentry-mac
```

Confirm that `pass` is available:

```
pass --version
# Expected: pass 1.7.4 or later
gpg --version
# Expected: gpg (GnuPG) 2.4.x
```

5 Configuration

5.1 Step 1: Generate a GPG Key

`pass` requires a GPG key to encrypt and decrypt entries. Generate one interactively:

```
gpg --full-gen-key
```

At the prompts, select:

- **Key type:** ECC (sign and encrypt, the default in GnuPG 2.4+). Ed25519 / Curve 25519 is the correct choice for new keys in 2026: constant-time operations, smaller keys, and equivalent or stronger security than RSA 4096. RSA remains an option but is not recommended for new keys.
- **Curve:** Curve 25519 (default when ECC is selected)
- **Expiration:** 2y (two years) rather than 0. A key that never expires remains valid indefinitely if compromised; a two-year expiration creates a forced review point. Add the renewal date to a calendar immediately. See Gotcha 3 for what happens when a key lapses.
- **Real name:** your name as you want it associated with the key
- **Email address:** the address you will use to identify the key when initialising the store
- **Passphrase:** a minimum of six random words (diceware) or a 25-character random string. Avoid sentences constructed from real phrases: dictionary attacks against GPG keys use pattern-aware wordlists. The passphrase protects every credential in the store.

After generation, verify the key is visible:

```
gpg --list-keys
```

The output will show a block beginning with `pub`, followed by the key fingerprint. Note the email address; it will be used in the next step.

5.2 Step 2: Initialise the Password Store

Initialise `pass` with the email address associated with the GPG key:

```
pass init your@email.address
```

This creates `~/.password-store/` and writes a `.gpg-id` file containing your email. Every subsequent `pass insert` call will encrypt to this identity.

5.3 Step 3: Insert and Retrieve a Credential

Insert a password by name:

```
pass insert Microsoft
```

`pass` will prompt for the password (and confirmation) and write the encrypted file to `~/.password-store/Microsoft.gpg`.

To copy the password to the clipboard without printing it:

```
pass -c Microsoft
```

The clipboard is cleared automatically after 45 seconds. To print to standard output instead (use with caution in shared terminal sessions):

```
pass Microsoft
```

Entries can be organised into subdirectories:

```
pass insert email/gmail
pass insert finance/chase
pass ls
```

`pass ls` displays the full tree of stored entries.

5.4 Step 4: Initialise git Tracking

`pass` can manage a git repository inside the store automatically. Every `pass insert`, `pass edit`, and `pass rm` is then committed without any extra steps.

```
pass git init
```

Use `pass git log` to review history. To push to a remote (for off-site backup), add one now:

```
pass git remote add origin git@your-private-host:password-store.git
pass git push -u origin main
```

Choose the remote carefully. The post-64 framework (see [Migrating Off Dropbox](#)) treats credentials as outside the sync layers that project content uses. The password store should not go to a public cloud git host, even in a ‘private’ repository: the entry names are metadata (they reveal which services you have accounts with) and a breach of the host exposes that metadata and the ciphertext. Prefer one of:

- A self-hosted Gitea or Forgejo instance on a home server or NAS.
- A VPS you control, accessed over SSH.
- No remote at all, with periodic offline exports of the GPG key and store as the backup strategy (see [Uninstall / Rollback](#)).

Accepting per-machine state for the password store (no remote sync, manual key import on each new machine) is architecturally honest and the simplest option for single-laptop setups.

Warning

Do not use GitHub, GitLab.com, or any third-party-hosted git service as the remote for your password store. Even a ‘private’ repository on those services places your credential metadata and ciphertext under a third party’s access controls.

5.5 Step 5: Guard the Store Against Accidental git Commits

Post 65’s sensitive-files inventory (Blocker 3) lists `~/.password-store` explicitly alongside `~/.gnupg`, `~/.ssh`, and `~/.aws` as items that ‘must never be committed to git, even in a private repository.’ Write the `.gitignore` entry before the first `git add` in any dotfiles repository; after the first commit the only remediation is a full `git filter-repo` rewrite and credential rotation.

For any dotfiles repository in use, add `.password-store` and `.gnupg` to its `.gitignore` before `git init`:

```
# In ~/.dotfiles/.gitignore, before git init:
.password-store
.gnupg
.ssh
.aws
secrets/*
!secrets/README.md
```

This step was missed during the original migration. Recovery after an accidental commit to a local-only repository:


```
git rm --cached -r '.password-store'  
git commit -m 'removed .pass'
```

If the data reached a public remote, `git rm --cached` is insufficient: use `git filter-repo` to rewrite history and rotate any credentials the ciphertext could represent. See [git-filter-repo](#) and post 65's Limitations section for the full procedure.

The encrypted files are GPG-protected and would require the private key and passphrase to decrypt, but publishing them publicly is still inadvisable: the entry names reveal which services you have accounts with and the ciphertext is material for offline cryptanalysis.



Figure 3: A terminal window showing the pass tree, with a `.gitignore` file open in a neighbouring split.

6 Migration from 1Password

`pass` ships with a collection of import scripts for common password managers. For 1Password, use the `1password2pass.rb` Ruby script.

Step 1: Export from 1Password. In the 1Password desktop app, choose *File > Export* and export in CSV format. Rename the file for convenience:

```
mv '1Password 2021-03-14, 04_47 PM (648 items and 1 folders).csv' \  
onepass.csv
```

Step 2: Download and run the conversion script:

```
# Obtain the script from the pass import collection
# (see https://www.passwordstore.org/#other)
chmod +x 1password2pass.rb
./1password2pass.rb onepass.csv
```

The script creates one `.gpg` file per entry in `~/.password-store/`, preserving folder structure where present. A 648-entry vault typically completes in under two minutes.

Step 3: Verify a sample entry:

```
pass ls | head -20
pass -c Amazon
```

Warning

Delete the plaintext CSV export after migration. It contains every password in unencrypted form. On macOS, `rm` does not overwrite file content on APFS/SSD; the data persists until the block is reused. If FileVault full-disk encryption is enabled (it should be), the block is protected at rest. If FileVault is not enabled, use a secure-delete tool before discarding the machine. In either case, ensure `~/Downloads/` is not synced to iCloud Drive before running the export.

7 Verification

After installation and initial population, run the following checks:

```
# 1. List all entries (confirms store is readable)
pass ls

# 2. Insert a test entry and retrieve it
pass insert test/smoke-test
pass -c test/smoke-test # copies to clipboard

# 3. Confirm GPG identity is correct
cat ~/.password-store/.gpg-id
```

If `pass ls` renders a tree and `pass -c` copies to the clipboard without a decryption error, the setup is correct.

8 Daily Workflow

Command	Action
<code>pass ls</code>	List all entries
<code>pass insert name</code>	Insert a new credential

Command	Action
<code>pass insert -m name</code>	Insert a multi-line entry (notes, TOTP)
<code>pass -c name</code>	Copy password to clipboard (clears in 45s)
<code>pass generate name 20</code>	Generate a 20-character random password
<code>pass edit name</code>	Open entry in <code>\$EDITOR</code>
<code>pass rm name</code>	Delete an entry
<code>pass mv old new</code>	Rename or move an entry
<code>pass git log</code>	Show the git history of the store
<code>gpg --list-keys</code>	Confirm your key is still present

After a short adjustment period, these commands displace the muscle memory built up around a GUI password manager. The clipboard-copy workflow (`pass -c name`) integrates cleanly with browser login fields and terminal credential prompts alike.

8.1 New-Laptop Bootstrap Position

In the three-layer bootstrap sequence from [post 64](#), credentials must be available before any credential-dependent step (IMAP sync, cloud storage auth, API key injection) can run. The pass setup therefore belongs immediately after the dotfiles layer, before any service authentication:

1. `git clone git@github.com:rgt47/dotfiles ~/dotfiles`
2. `cd ~/dotfiles && ./install.sh`
Installer prints: "Restore ~/.aws, ~/.ssh, ~/.gnupg
from secure storage"
3. `gpg --import private-key.asc` # from offline backup
4. `pass git clone git@your-host:pass.git ~/.password-store`
5. `pass ls` # verify store is readable
6. `mbsync -a` # now IMAP credentials exist
7. ... (remaining bootstrap steps)

Step 2 is the `install.sh` from [post 65](#), which deploys dotfiles symlinks and prints a reminder to restore `~/.gnupg` before any credential-dependent step runs. The offline GPG key backup (`gpg --export-secret-keys --armor ...`) is a prerequisite for step 3 on any new machine; keep it on encrypted removable media, not in cloud storage.

9 Things to Watch Out For

1. **The `.password-store` directory must not appear in any public git repository.** Even though files are GPG-encrypted, their names reveal which services you have accounts with. Add `.password-store` to `.gitignore` in your dotfiles repo before doing anything else. This step was missed during the original migration and required a history rewrite to correct.

2. **GPG passphrase prompts can fail silently on macOS without pinentry-mac.** If `pass insert` appears to succeed but subsequent `pass` commands fail with a decryption error, the GPG agent may not have cached the passphrase correctly. Install `pinentry-mac` and add this to `~/.gnupg/gpg-agent.conf`:

```
pinentry-program /opt/homebrew/bin/pinentry-mac
default-cache-ttl 300
max-cache-ttl 600
```

The path `/opt/homebrew/bin/pinentry-mac` is correct for Apple Silicon. On Intel Macs, Homebrew installs to `/usr/local/bin/`; substitute accordingly, or use `$(brew --prefix)/bin/pinentry-mac` in a setup script. The `default-cache-ttl` (300 s) controls how long after the last use the passphrase stays cached; `max-cache-ttl` (600 s) is the absolute ceiling. Omitting these leaves the default at 600 s, which is long for an unattended machine.

3. **The GPG key expiration date applies to all stored credentials.** A key set to expire that is not renewed will make every entry unreadable. Either set no expiration or establish a calendar reminder to extend the validity date before it lapses.
4. **`pass -c` clears the clipboard after 45 seconds, but clipboard managers bypass this entirely.** On macOS this requires `pbcopy`, which is present by default. On Linux it requires `xclip` or `xsel`. If neither is available, `pass -c` will fail. More importantly: any clipboard manager (Alfred, Raycast, ClipMenu, the macOS Universal Clipboard synced via iCloud) captures the `pass -c` output before the 45-second timer fires and stores it permanently, possibly syncing it to other devices. Disable clipboard history, or exclude `pass` invocations, in any clipboard manager running on the system. On a two-laptop setup with iCloud Universal Clipboard enabled, every `pass -c` on one machine appears in plaintext on the other.
5. **The `1password2pass.rb` migration script handles standard CSV exports but may mangle multi-line secure notes or custom field names.** Review a sample of migrated entries for completeness before deleting the source vault.
6. **The store is as secure as your GPG private key.** If the private key is stored unencrypted, or if the passphrase is weak, the GPG layer provides limited protection. Use a passphrase of at least 20 characters.
7. **Renaming the GPG key email address after store initialisation requires re-encrypting every entry.** The `.gpg-id` file stores the recipient identity; changing it with `pass init new@email` triggers a batch re-encryption of the whole store.
8. **Pass for iOS generates its own SSH key pair; add that key to your self-hosted server, not your desktop key.** A common mistake is copying the desktop SSH public key to the server's `authorized_keys` while the iOS app is using a different key it generated internally. The symptom is a 'permission denied (publickey)' error on every sync attempt. Go to the app's Settings, copy the displayed public key, and append it to `~/.ssh/authorized_keys` on the git server as a separate entry. Also verify that the GPG key fingerprint shown in the app matches the fingerprint in `~/.password-store/.gpg-id`.

10 Uninstall / Rollback

To remove `pass` and its configuration from the workstation:

```
# 1. Remove the password store (IRREVERSIBLE without a backup)
#    Only run this if you have exported or backed up all credentials.
rm -rf ~/.password-store

# 2. Remove the GPG key (IRREVERSIBLE)
gpg --list-keys          # note the key ID
gpg --delete-secret-and-public-key KEYID

# 3. Uninstall the tools
brew uninstall pass gnupg pinentry-mac
```

⚠ Warning

Back up the password store and the GPG private key before uninstalling. Export the private key with `gpg --export-secret-keys --armor your@email > key-backup.asc` and store the file offline.



Figure 4: A small lockbox closed on a wooden surface, symbolising credentials secured and at rest.

11 What Did We Learn?

11.1 Lessons Learnt

Conceptual Understanding:

- `pass` is thin by design: it delegates all cryptographic work to GPG and all version control to git, composing two well-audited tools rather than reimplementing them.
- A GPG identity is portable: the same key pair can encrypt and decrypt entries across any machine where the private key is installed, which is the correct model for multi-device credential access.
- Password store organisation mirrors filesystem organisation: the mental model for `pass insert email/gmail` is identical to `mkdir email && touch email/gmail`, which makes navigation intuitive for anyone comfortable at a terminal.
- Accidental publication of encrypted data is still a security event: ciphertext reveals metadata (entry names, creation dates) and provides material for offline cryptanalysis.

Technical Skills:

- Generating a GPG key with `gpg --full-gen-key` and inspecting it with `gpg --list-keys` are foundational skills that transfer to signing git commits, verifying software releases, and encrypting arbitrary files.
- The `pass git` subcommand proxies any git command against the password store repository, making it easy to push the store to a private remote for backup.
- `git rm --cached -r` removes a tracked directory from the index without deleting the local copy, which is the correct recovery command when a sensitive directory is accidentally staged.
- `pinentry-mac` bridges the macOS graphical environment and the GPG agent; configuring it correctly eliminates most passphrase prompt failures.

Gotchas and Pitfalls:

- Forgetting to add `.password-store` to `.gitignore` before the first `git add .` in a dotfiles repo is the most common first-week mistake.
- A GPG key that has expired will cause all `pass` operations to fail with a non-obvious error message; check `gpg --list-keys` first when debugging decryption failures.
- The 1Password CSV export may contain entries whose names include characters that are invalid in filenames (`/`, `:`, `?`); the migration script silently truncates or skips these.
- The clipboard-clear timer (45s) runs in a background process; it will not fire if the shell exits before the timer elapses.

11.2 Limitations

- **Multi-device synchronisation is manual.** `pass` can push and pull from a git remote, but the user must manage this explicitly; there is no background sync daemon.
- **Mobile access requires a companion app and key transfer.** Pass for iOS exists and syncs via a git remote, but it requires importing the GPG private key to the device. The key transfer itself (AirDrop or similar) is a security event: the private key is briefly in plaintext and in transit. See Appendix A for the full procedure.

- **No breach monitoring.** Unlike some commercial managers, `pass` does not check credentials against known-breached password lists.
- **Bulk password generation is not built in.** Generating unique passwords for 600 migrated entries is a manual process; the migration script imports existing passwords but does not regenerate them.
- **The security model depends entirely on the GPG key.** Loss of the private key, or compromise of the passphrase, results in loss of access to all credentials. Offline key backups are essential but add operational complexity.

11.3 Opportunities for Improvement

1. Push the store to a private self-hosted remote (Gitea, Forgejo, or a bare git repository on a NAS) using `pass git init` and `pass git remote add` (covered in Configuration Step 4). Avoid third-party cloud git hosts even with private repositories.
2. Install and configure Pass for iOS to enable mobile access; see Appendix A for the git sync, GPG key export, and SSH configuration procedure.
3. Configure the GPG agent to use a hardware security key (YubiKey) so the private key never exists in plaintext on disk.
4. Add a shell alias (e.g., `alias p='pass -c'`) to reduce the keystrokes for the most common operation.
5. Periodically audit the store with `pass ls` to remove stale entries for defunct accounts and services.
6. Enable `pass git push` as a post-commit hook so the remote backup is updated automatically after every change.

12 Wrapping Up

Switching to `pass` from a combination of browser password managers and a commercial vault took one afternoon and produced a credential store that is simpler, more auditable, and more portable than anything it replaced. The migration of 648 entries from 1Password was completed in under two minutes using the `1password2pass.rb` script. The entire store is a directory of GPG-encrypted files: no proprietary format, no subscription required to read it.

The most instructive moment in the setup was the accidental commit to a public repository. Recovering from it was straightforward, but the incident made concrete a rule that had previously felt theoretical: sensitive directories belong in `.gitignore` before the repository is created, not after.

In conclusion, four points merit emphasis. First, `pass` stores passwords as GPG-encrypted files in a plain directory tree, composing GPG and git rather than replacing them. Second, the store must be added to `.gitignore` in any dotfiles repository before the first commit; this step is easy to miss and non-trivial to reverse. Third, `pinentry-mac` is required on macOS for reliable passphrase prompts outside the terminal and must be configured in `~/.gnupg/gpg-agent.conf`. Fourth, the `1password2pass.rb` script migrates a 1Password CSV export to the store in one command; the plaintext CSV should be deleted afterwards.

13 See Also

Official documentation:

- passwordstore.org: the official `pass` homepage with install instructions and the full list of compatible import scripts
- passwordstore.org/#other: companion apps and import scripts, including `1password2pass.rb`
- [GnuPG documentation](#)

Related posts:

- Post 64, [Migrating Off Dropbox: Beyond Dotfiles](#): the three-layer framework that defines where the password store belongs in a multi-laptop workflow and why cloud-mounted git repos are structurally wrong.
- Post 65, [Multi-Laptop macOS Bootstrap](#): the concrete implementation of post 64's framework: the `install.sh` that deploys dotfiles and prints the 'restore `~/.gnupg` from secure storage' reminder that step 3 of this post's bootstrap sequence depends on. Also lists `~/.password-store` explicitly in its sensitive-files inventory.
- Post 24, [Creating a GitHub Dotfiles Repository](#): the architectural design that post 65 implements and that this post's bootstrap sequence builds on.
- [Setting Up Git on a New Machine](#): relevant if you plan to track the password store in a private self-hosted git remote.

14 Reproducibility

Tested configuration:

Component	Version
Operating system	macOS 15.4 (Sequoia)
pass	1.7.4
gnupg	2.4.x
pinentry-mac	1.3.x
Shell	zsh 5.9
Homebrew	4.x
Last verified	2026-05-17

Configuration files:

- `analysis/configs/gpg-agent.conf`: `pinentry-mac` configuration
- `analysis/configs/gitignore-guard.sh`: one-liner to add `.password-store` to an existing dotfiles `.gitignore`

To reproduce the core setup:

```
brew install pass gnupg pinentry-mac
gpg --full-gen-key          # follow prompts
pass init your@email.com
pass insert test/smoke
pass -c test/smoke         # confirm clipboard copy works
```

15 Appendix A: iOS Sync with Pass for iOS

The Pass for iOS app reads the same GPG-encrypted `.gpg` files that the desktop `pass` command produces. Once the store is in a private git remote (Configuration Step 4), the iOS app can pull and decrypt entries on the device. The procedure has four parts.

15.1 A.1 Confirm the Remote is Set Up

From the desktop:

```
pass git remote -v
# Should show: origin  git@your-private-host:password-store.git
pass git push
```

If no remote is configured, add one first (see Configuration Step 4). Note that this remote should be a self-hosted server, not a public cloud service. The iOS app will pull directly from the same remote.

15.2 A.2 Export the GPG Key

The app needs your private key to decrypt entries. Export both the public and private key to armored text files:

```
gpg --export-armor YOUR_KEY_ID > public.asc
gpg --export-secret-keys --armor YOUR_KEY_ID > private.asc
```

Transfer `private.asc` to the iPhone securely. AirDrop is acceptable for this step: the file is protected only by your GPG passphrase, so ensure the passphrase is strong before transferring. Delete the `.asc` files from the desktop immediately after transfer.

Warning

The exported private key is a sensitive file. Do not email it, do not store it in Dropbox or iCloud unencrypted, and delete it from the Files app on the iPhone after importing into Pass.

15.3 A.3 Configure the iOS App

Install **Pass - Password Store** from the App Store. In the app:

1. Go to **Settings** → **Password Store**.
2. Set the **Git Repository URL** to your self-hosted remote.

SSH (recommended for a self-hosted remote):

- The app generates its own SSH key pair. Go to **Settings** → **SSH Key**, copy the displayed public key, and add it to `~/.ssh/authorized_keys` on your self-hosted git server. Do not copy your desktop key; the iOS app generates a separate key internally.

HTTPS (if your remote supports it):

- Use the credentials for your self-hosted service (Gitea, Forgejo, etc.). If the service supports tokens, use a token scoped to the password-store repository only rather than your account password.

15.4 A.4 Import the GPG Key and Sync

1. In the app, go to **Settings** → **PGP Key** → **Import Key**.
2. Select the `private.asc` file from Files.
3. Enter your GPG passphrase when prompted.
4. Tap **Sync**. The app pulls the repository and decrypts the entry list.

Confirm that a known entry is visible and tappable. Tapping an entry copies the password to the iOS clipboard.

15.5 A.5 Common Snags

- **‘Permission denied (publickey)’ on sync:** the app’s SSH key has not been added to your self-hosted server’s `authorized_keys`. Copy the public key from **Settings** → **SSH Key** in the app and append it to `~/.ssh/authorized_keys` on the git server.
- **‘No secret key’ decryption error:** the fingerprint in the app does not match the `.gpg-id` in the repository. Check both values: `cat ~/.password-store/.gpg-id` on the desktop, **Settings** → **PGP Key** in the app.
- **HTTPS 401 error:** the Personal Access Token has expired or was not granted write access to the repository. Regenerate it.

16 Appendix B: Security Considerations

The following items do not fit neatly into the setup steps but materially affect the security posture of the overall credential architecture. Post 67 addresses these topics in depth; this appendix summarises the key points for quick reference.

16.1 B.1 GPG Subkey Architecture

The setup above creates a single key pair used for both certification and encryption. The correct operational model for daily use is:

- **Master key (certify only):** signs new subkeys and revocation certificates. Generated once, then moved to offline storage or a hardware security key (YubiKey). Never used in daily operation.
- **Encryption subkey:** the only key needed by `pass` for day-to-day encrypt/decrypt operations. Shorter expiry than the master.
- **Signing subkey (optional):** used for `git commit --gpg-sign`.

With this architecture, the master private key never needs to exist on the laptop's disk during normal operation. A compromised laptop exposes only the encryption subkey, which can be revoked by signing a new revocation certificate with the (offline) master key. This is the mechanism by which hardware security keys (Opportunity #3) work. Implementing the full subkey architecture is beyond this post's scope but is the recommended next step after the basic setup is stable.

16.2 B.2 AWS and API Credentials via pass

The `~/.aws/credentials` file is plaintext on disk. Having a working `pass` store makes it straightforward to remove it:

```
# Store AWS credentials in pass
pass insert aws/access-key-id
pass insert aws/secret-access-key

# Retrieve on demand via a shell function in ~/.zshrc:
aws_env() {
    export AWS_ACCESS_KEY_ID=$(pass aws/access-key-id)
    export AWS_SECRET_ACCESS_KEY=$(pass aws/secret-access-key)
}
```

Call `aws_env` at the start of a session that needs AWS access. The plaintext credentials are in memory only, not on disk. Remove `~/.aws/credentials` after confirming the function works. The same pattern applies to any plaintext credential file (`~/.npmrc` tokens, `~/.config/rclone/rclone.conf`, API keys in `~/.env`).

16.3 B.3 FileVault

Full-disk encryption (FileVault on macOS) is a prerequisite, not an enhancement, for any local credential store. The GPG encrypted files in `~/.password-store/` are strong, but the GPG agent caches the decrypted private key in memory; if the machine is lost while unlocked, or if a swap file captures memory, the cached key is readable without FileVault. Confirm FileVault status:

```
fdesetup status
# Expected: FileVault is On.
```

If FileVault is not enabled, enable it before populating the password store.

16.4 B.4 SSH Key Policy for Multi-Laptop Setups

Post 65's bootstrap sequence says 'Restore ~/.ssh from secure storage.' For SSH private keys, the correct multi-machine policy is NOT to copy keys between machines but to generate a new key pair on each machine:

```
ssh-keygen -t ed25519 -C "$(hostname) - $(date +%Y%m) "
```

Then add the new public key to GitHub, servers, and anywhere the previous key was authorised. This is more work initially, but it means a compromised laptop exposes only that machine's key, and revocation is surgical. Copying a private key between machines means all machines are compromised if any one of them is.

The one exception is the GPG private key, which is a single identity and must be the same across machines by design. The mitigant is the subkey architecture described above, which limits the blast radius of any single machine compromise.

17 Let's Connect

Have questions, suggestions, or spot an error? Let me know.

- **GitHub:** [rgt47](#)
- **Twitter/X:** [@rgt47](#)
- **LinkedIn:** [Ronald Glenn Thomas](#)

I would enjoy hearing from you if:

- You spot an error or a better approach to any of the steps above.
- You have suggestions for topics you would like to see covered.
- You use a different OS or password manager and want to compare notes.

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17.1 Related posts in this cluster

This post is part of the *Security, Backup, and Sync* series. Recommended reading order:

1. Post 31: [Research Backup Architecture](#)
2. Post 32: [Migrating Off Dropbox: Beyond Dotfiles](#)
3. **Post 33: Setting Up pass: a Unix Password Manager** (this post)
4. Post 34: [Secrets Management for the Workflow Construct](#)
5. Post 35: [Security Foundations for a Multi-Laptop Research Cluster](#)